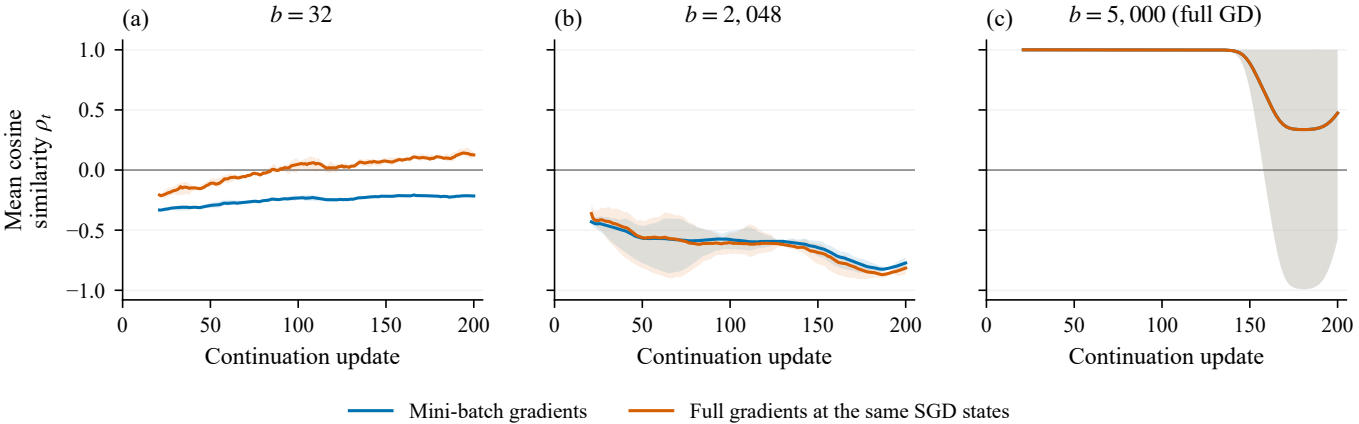

(a)
$b = 32$
(b)
$b = 2,048$
(c)
$b = 5,000$ (full GD)
Mean cosine similarity ρ_t
1.0
0.5
0.0
−0.5
−1.0
0
50
100
150
200
Continuation update
Mini-batch gradients
Full gradients at the same SGD states